Albatross: a 7-week deployment playbook, run on a real marketplace dataset

OVERVIEW

From first data drop to a go/no-go readiness score, a deployment-health dashboard and an A/B readout a customer can trust.

REAL DATASET · PAGE 3

73/100

Readiness of the first data drop: **go with conditions**. Train now; fix the 9.3% of events on unknown items and session tracking before the A/B test.

REAL DATASET · PAGE 4

0.1%

Recall@20 of item-to-item co-visitation when the next item is new, against 36% on established items. That gap is the case for content-aware, session-aware models.

REAL DATASET · PAGE 5

±14%

Smallest purchase lift a 3-week test can detect at this dataset's traffic (±2% for engagement). Engagement must be the primary metric.

SIMULATED · PAGES 7-8

+2.1%

Conversion lift needed to cover the list price. The simulated test reads +3.3% [+0.1%, +6.6%]: go, with a holdout.

Five questions a Deployment Strategist answers

1 · **Product** What does Albatross sell, and how does an evaluation run? (p.1)

2 · **Intake** Is the customer's first data drop usable, and what must be cleaned? (p.2)

3 · **Readiness** Go, Go with conditions, or Not yet for the first training run? (p.3)

4 · **Baselines** Where does the incumbent-style logic fail, and why does that matter? (p.4)

5 · **Live test** How is the A/B test designed, is the deployment healthy, did it win, does it pay? (p.5-9)

Recommendation: start training on the cleaned drop, send the customer 5 written conditions, run a 3-week test with engagement as the primary metric, and move to production with a 5 % holdout so revenue per visitor is confirmed before the first renewal talk.

Data & methods

PUBLIC DATA Albatross's own pages (product, 7-week evaluation, customer results, pricing), collected 2026-11-11.

REAL DATASET RetailRocket: 2.76 M real e-commerce events + catalog, used *as if* it were a new customer's first data drop.

SIMULATED live health series and A/B test (weeks 5-7), calibrated on the real base rates, effects kept below published lifts.

Tools SQL in DuckDB (checks, sessions, co-visitation), Python (power, z-tests, bootstrap, chi-square), Plotly dashboard, a reusable adequacy toolkit with tests.

How to read this report

Titles state the conclusion of the page. **Tags** PUBLIC DATA / REAL DATASET / SIMULATED mark where the numbers come from. **Colours** blue = data or treatment, grey = comparison or control, red = alert, green = go; dashed lines are thresholds. **Albatross's models are proprietary:** the simple baselines here are not theirs and do not try to reproduce them.

Sources: data/SOURCES.md · Independent case study based on public data · not affiliated with or endorsed by Albatross AI

Albatross sells a measured outcome: live on the homepage in 7 weeks, judged by an A/B test

PUBLIC DATA

Perception models update on every click and make new listings recommendable within minutes. Customers publish lifts of +84 % to +119 % in homepage engagement; the 7-week evaluation is where that proof is produced.

What Albatross sells

Perception models: sequential (transformer) embedding models that read each user's live sequence of events (views, clicks, searches, purchases) and update on every click.

- **Catalog layer**: content embeddings (text, images) per item, so a new listing is recommendable *within minutes, before its first click*.
- **User layer**: a real-time user embedding from the interaction sequence.
- **Serving**: returns the k best items through three APIs (Catalog, Event, Prediction).

Four primitives, one API: Discovery Feed (user_id, k) · Similar Items (item_id, k) · Playlist (item_ids, k) · Contextual Search (query, user_id, k).

Research: SwapRec (RecSys 2026) and DenseRec on cold / brand-new items; content-enriched sequential recommenders; Multinomial Blending for promotions.

Announced scale

3B+
requests / month

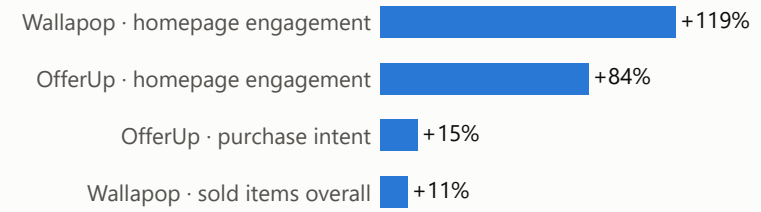
50M+
users / month

150M+
products orchestrated

12
countries with live marketplaces

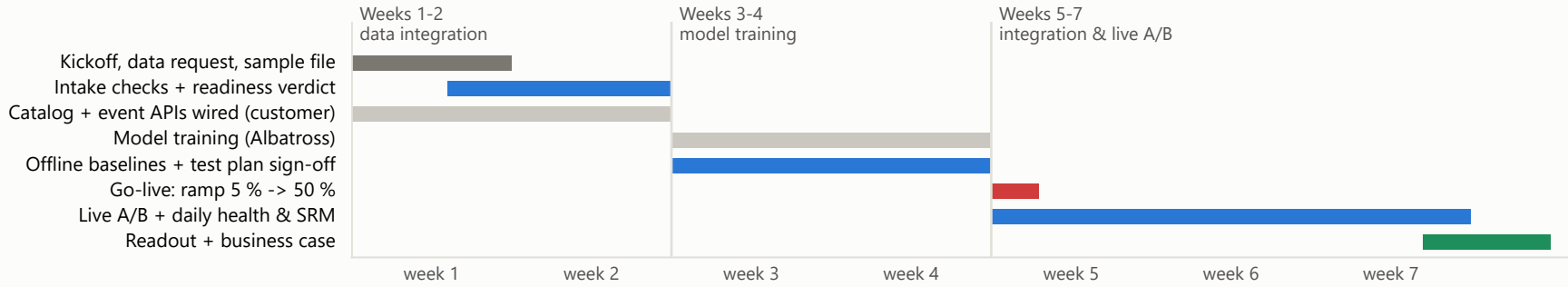
Founded 2024, Switzerland; $16M seed (MMC Ventures, Redalpine, Daphni); Tech for Retail 2025 award. Public pricing: platform fee from €10k to €40k / month by MAU tier + €0.10-0.50 per 1,000 predictions.

Results published by customers (as shown on usealbatross.ai)



Wallapop's case study also reports similar-items catalog coverage rising from 50 % to 94 % and +49.9 % conversion to sold item. OfferUp: Discovery Feed A/B-tested on the homepage against the incumbent system.

The 7-week evaluation: the Deployment Strategist's job, week by week



Customer side: one engineering lead + one PM. Data: event triplets (user ID, event type, item ID, timestamp) + catalog access; clicks only is enough to start. Fixed 3-month evaluation, no auto-renewal; production is a separate agreement. Blue bars = the Deployment Strategist's deliverables in this study.

Sources: usealbatross.ai (home, technology, evaluation, primitives, customers, pricing, company), collected 2026-11-11; data/SOURCES.md · Independent case study based on public data · not affiliated with or endorsed by Albatross AI

The first drop is usable, but 9.3% of events point to items missing from the catalog

REAL DATASET

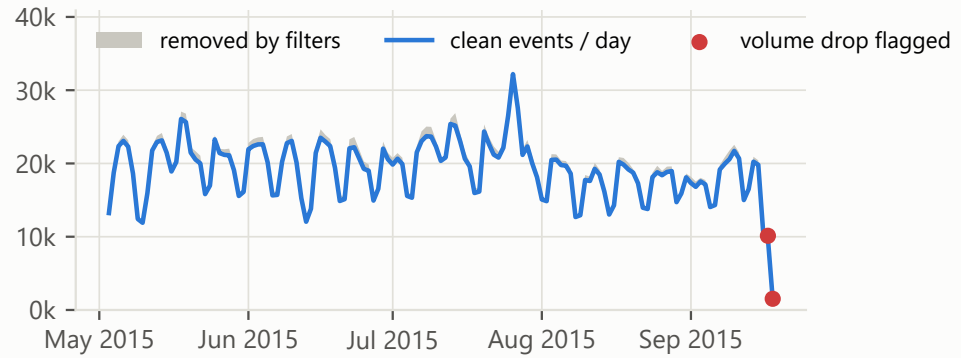
17 automated checks on the customer's first drop (sql/01_intake_checks.sql): no blocking issue; the warnings are a catalog export gap, partial last days, 3.4% of events from bot-like users, and 28% of add-to-carts with no earlier view.

11 checks OK · 6 warnings · 0 blocking

Status	Check	Rows	Share	What we do / ask
WARNING	Events on items missing from the catalog	255,585	9.3%	Ask for a full catalog export (including sold / delisted items).
WARNING	Catalog items with no event in the period	231,807	55.6%	Expected in a marketplace: these items depend on content embeddings (cold start).
WARNING	Days below 50 % of the previous 7-day median	2	1.4%	Check each date with the customer (outage, partial export, holiday).
WARNING	Events from users with >100 events in one day (bot rule 1)	93,012	3.4%	Exclude these users from training and from A/B metrics.
WARNING	Purchases with no earlier view of the item	2,633	11.7%	Some is normal (other device, e-mail link); a high share means views are missing.
WARNING	Add-to-carts with no earlier view of the item	19,188	27.7%	Same as above: check whether every surface sends view events.

Also checked and OK: schema, timestamps (Unix ms, UTC), nulls, event types, duplicate catalog ids, exact duplicates (460), repeats under 1 s (4,705), missing days (0), bursts.

Daily volume: the export ends mid-day



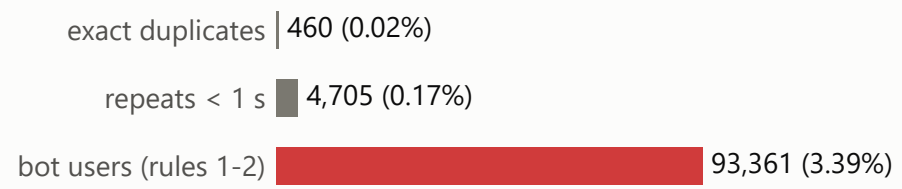
2015-09-17 (10128 vs median 20427), 2015-09-18 (1528 vs median 16859): the last two days are partial and must be dropped from daily KPIs.

Filtering rules, applied in this order (sql/01_intake_checks.sql)

#	Rule	Why
1	Drop rows missing a field or with a timestamp outside 2000-01-01 .. now	unusable or unit error
2	Keep one copy of exact duplicates (same user, item, type, timestamp)	tracker retries
3	Drop a repeat of the same user / item / type less than 1 s after the previous one	double taps, reloads
4	Drop every event of a user with >100 events in a day or 20+ in one minute	crawlers, scripts

Kept but flagged: events on unknown items (needed to measure the catalog gap) and purchases without a view (a tracking question for the customer, not a reason to delete sales).

Filtering rules remove 98,526 events (3.6%)



Bot rule 1: more than 100 events in a day (129 users flagged). Bot rule 2: 20+ events in one minute. Same logic as reconciling counted vs real clicks: bots, previews and double taps removed with written rules, applied identically to training data and A/B metrics.

Verdict: Go with conditions (73/100). Train now; fix the catalog feed and session tracking before the A/B

REAL DATASET

139 days of history and a fully categorised catalog pass. Short sessions, low daily volume (19,836 events/day) and a cold long tail are conditions, not blockers.

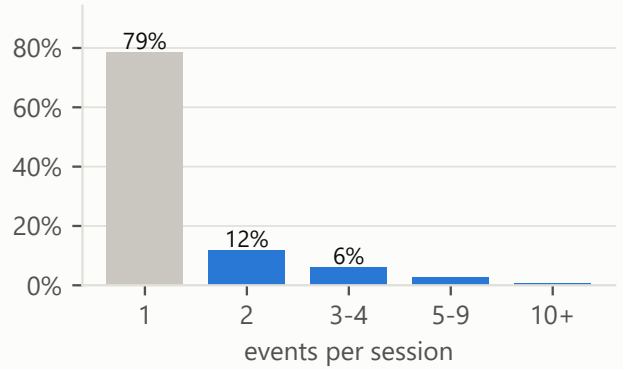
Readiness scorecard

GO WITH CONDITIONS · 73/100

Grade	Criterion (* critical)	Value	Rule
PASS	History length (days) *	139	pass >= 56; fail < 28
CONDITION	Median events per day	19,836	pass >= 50,000; fail < 5,000
CONDITION	Users with 2+ events	29%	pass >= 40%; fail < 20%
CONDITION	Sessions with 2+ events *	21%	pass >= 40%; fail < 20%
CONDITION	Events on items missing from catalog *	9.3%	pass <= 1.0%; fail >= 10.0%
PASS	Catalog items with a category	100%	pass >= 95%; fail < 70%
PASS	Events beyond views (cart, purchase)	2.9%	pass >= 1.0%; fail < 0.2%
CONDITION	Catalog items with < 5 interactions	81%	pass <= 50%; above: condition
PASS	Blocking intake checks *	0	pass = 0, else fail

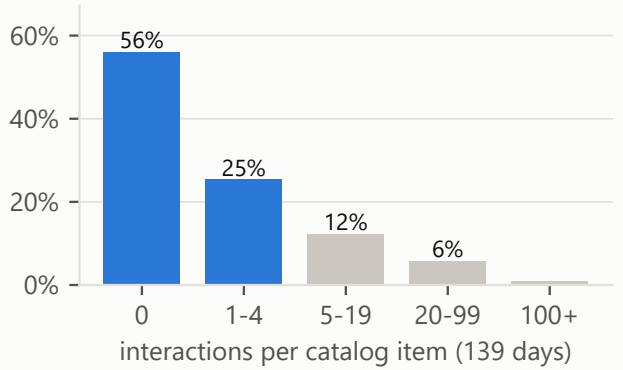
Not yet = a critical criterion fails. Go with conditions = no critical failure, some conditions. Thresholds are this playbook's rules (src/adequacy/readiness.py), not Albatross's.

Short sessions: 21% have 2+ events



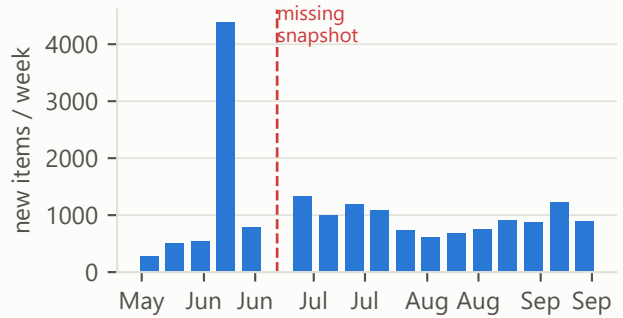
Median 1 event per session; 29% of users come back. In-session intent is the core signal: this is the first condition.

A cold catalog: 81% of items have under 5 interactions



Density of the user x item matrix: 6.4e-06. Top 1 % of items take 23% of events. Content embeddings are what serve this long tail.

~870 new catalog items per week



Weekly catalog snapshots; one snapshot is missing (2015-06-21): ask for a daily delta feed.

Conditions sent to the customer

- Add more surfaces or event types (clicks on all pages, not only the homepage).
- Send a stable user id (logged-in id or long-lived cookie) so visits link together.
- Check that every page view / click is tracked, not only the first one.
- Send a full catalog export, including sold and delisted items.
- Make sure every listing has a title, category and image URL: content is what makes the long tail and brand-new items recommendable.

Full e-mail generated by the toolkit: data/public/intake/customer_message.txt

Data: RetailRocket e-commerce dataset (public, CC BY-NC-SA 4.0), 2,756,101 events, May-Sep 2015 · code: sql/, src/adequacy/ · thresholds: src/adequacy/readiness.py · Independent case study based on public data · not affiliated with or endorsed by Albatross AI

Co-visitation finds the next item 31% of the time, but almost never when the item is new

REAL DATASET

Popularity is near useless for discovery (Recall@20 1.6%). Co-visitation is a strong incumbent on established items and collapses on the cold and new items that make up a marketplace's long tail.

Three incumbent-style baselines, time split

Train: 16 weeks to 2015-08-21 · Test: the next 4 weeks, 105,753 next-item predictions in 45,722 sessions. Task: the user is on item P, which different item comes next?

Baseline	Recall@20	NDCG@20	Coverage	Items shown
(a) Global popularity	0.7%	0.003	0.02%	20
(b) Recent popularity (7 days)	1.6%	0.006	0.06%	67
(c) Co-visitation + popularity fill	30.6%	0.169	57.82%	61,566

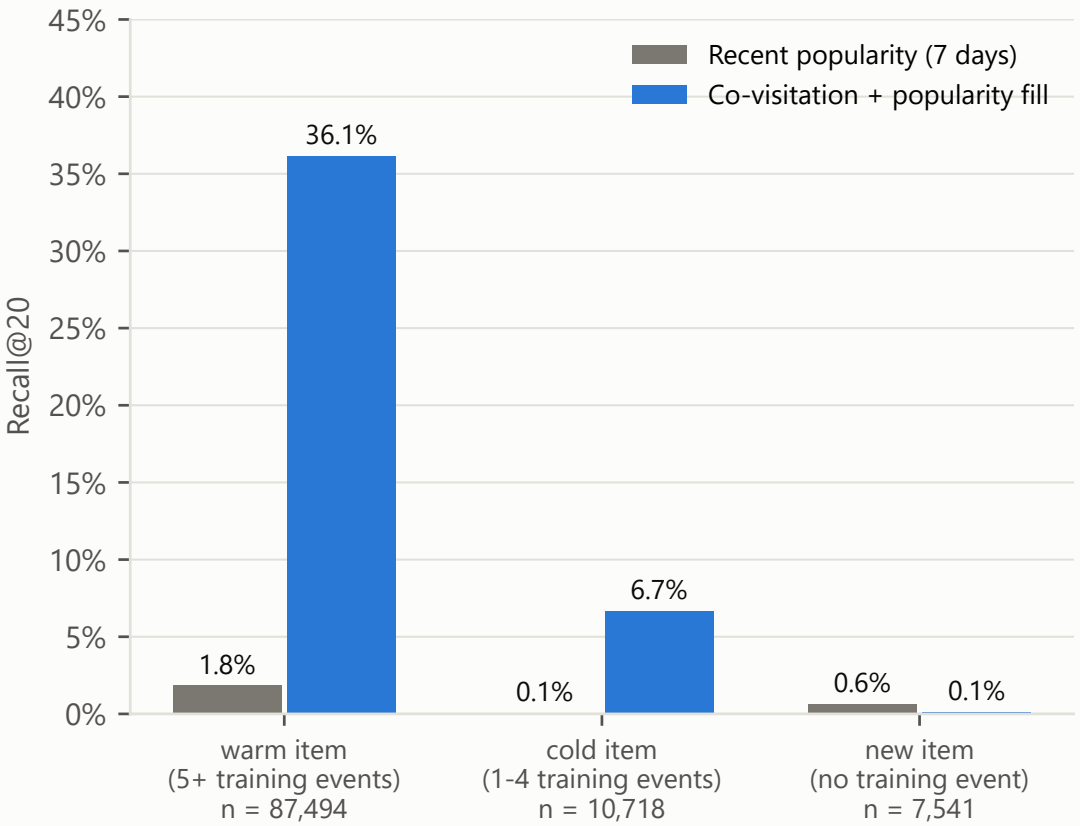
Coverage = share of the items people interacted with in the test weeks that the baseline ever recommends. Popularity shows the same 20-67 items to everyone.

Where the simple logic breaks, and what that says about the product

- **New items** (7.1% of next items had no training history): co-visitation recall 0.1%, against 36.1% on warm items. Content embeddings are built for exactly this.
- **Items with no co-visitation history** (12.7% of cases): recall falls to 0.8%, the popularity fallback.
- **Longer context is ignored**: co-visitation uses the last item only; recall is 28.4% with 2+ items seen vs 33.5% with 1. A sequence model reads the whole session.
- **Offline is not online**: these metrics rank candidates on logged data; only the live A/B (pages 5-7) measures value.

These baselines are not Albatross's models and do not estimate their performance.

Co-visitation works on established items, fails on new ones



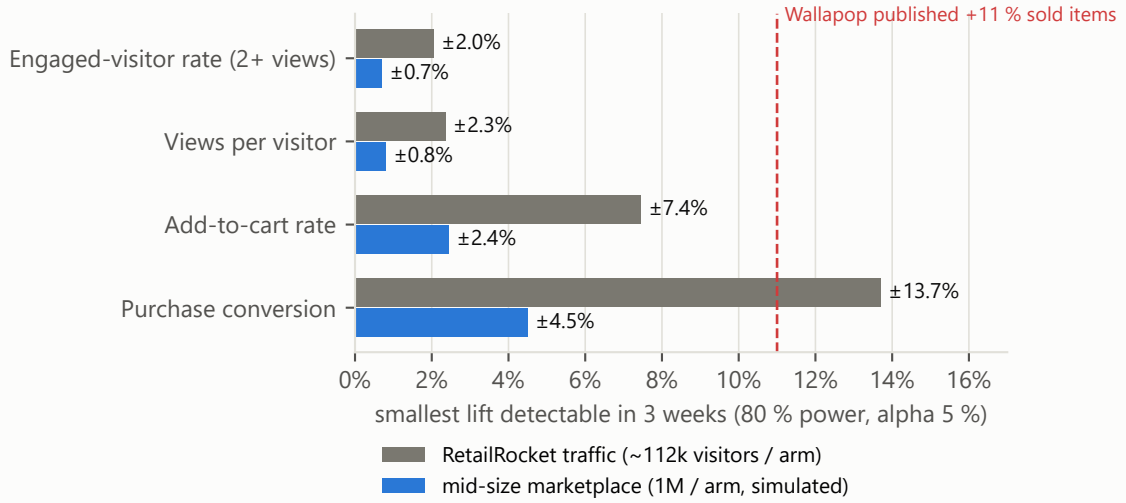
Data: RetailRocket e-commerce dataset (public, CC BY-NC-SA 4.0), 2,756,101 events, May-Sep 2015 · code: sql/, src/adequacy/ · sql/03_baselines.sql, src/baselines.py · Independent case study based on public data · not affiliated with or endorsed by Albatross AI

At this dataset's traffic, 3 weeks can read engagement (±2%) but not purchases (±14%)

REAL DATASET

The test is sized on the customer's real base rates before go-live, so the primary metric is one the test can actually answer, and every rule for reading it is written down in advance.

What a 3-week test can detect



Real base rates per visitor over 3-week windows: engaged 25.9%, add-to-cart 2.59%, purchase 0.80%, 1.72 views (sd 3.38).

Metric	Lift to detect	Visitors / arm	Weeks, RetailRocket traffic	Weeks, simulated customer
Engaged-visitor rate	+2%	112,813	4	1
Engaged-visitor rate	+5%	18,221	1	1
Engaged-visitor rate	+10%	4,625	1	1
Engaged-visitor rate	+20%	1,189	1	1
Purchase conversion	+2%	4,918,724	> 8	> 8
Purchase conversion	+5%	798,587	> 8	3
Purchase conversion	+10%	204,474	6	1
Purchase conversion	+20%	53,531	2	1

Analysis plan, signed before go-live

Primary	Homepage feed clicks per visitor (engagement), the metric customers publish and the one this traffic can read.
Secondary	Engaged-visitor rate, add-to-cart rate, purchase conversion, revenue per visitor.
Guardrails	Page load time (max +50 ms), bounce rate (max +0.5 pt), error and fallback rates.
Unit	Visitor, sticky hash of user_id + salt, 50/50, same assignment on web and apps.
Duration	21 days fixed (3 full weekly cycles = weeks 5-7). No decision before day 21: no peeking.
Stop early	Only for a guardrail breach or a SEV1 incident.
SRM	Chi-square on the split daily and by device; alarm at $p < 0.001$; investigate before reading any result.
Tests	Welch z-test + bootstrap CI (visitor resampling); Holm correction across segments; novelty read week by week.

Sample size, the formula behind the chart

$$n \text{ per arm} = (z_{\alpha/2}\cdot\sqrt{(2\bar{p}\bar{q})} + z_{\beta}\cdot\sqrt{(p_1q_1+p_2q_2)})^2 \,/\, (p_2-p_1)^2$$

α = 5 % two-sided, power = 80 %

Detecting +5 % on purchase conversion needs 798,587 visitors per arm. RetailRocket reaches ~111,661 per arm in 3 weeks: purchases are out of reach, engagement is not. The simulated customer (1M per arm) can read ±4.5% on conversion.

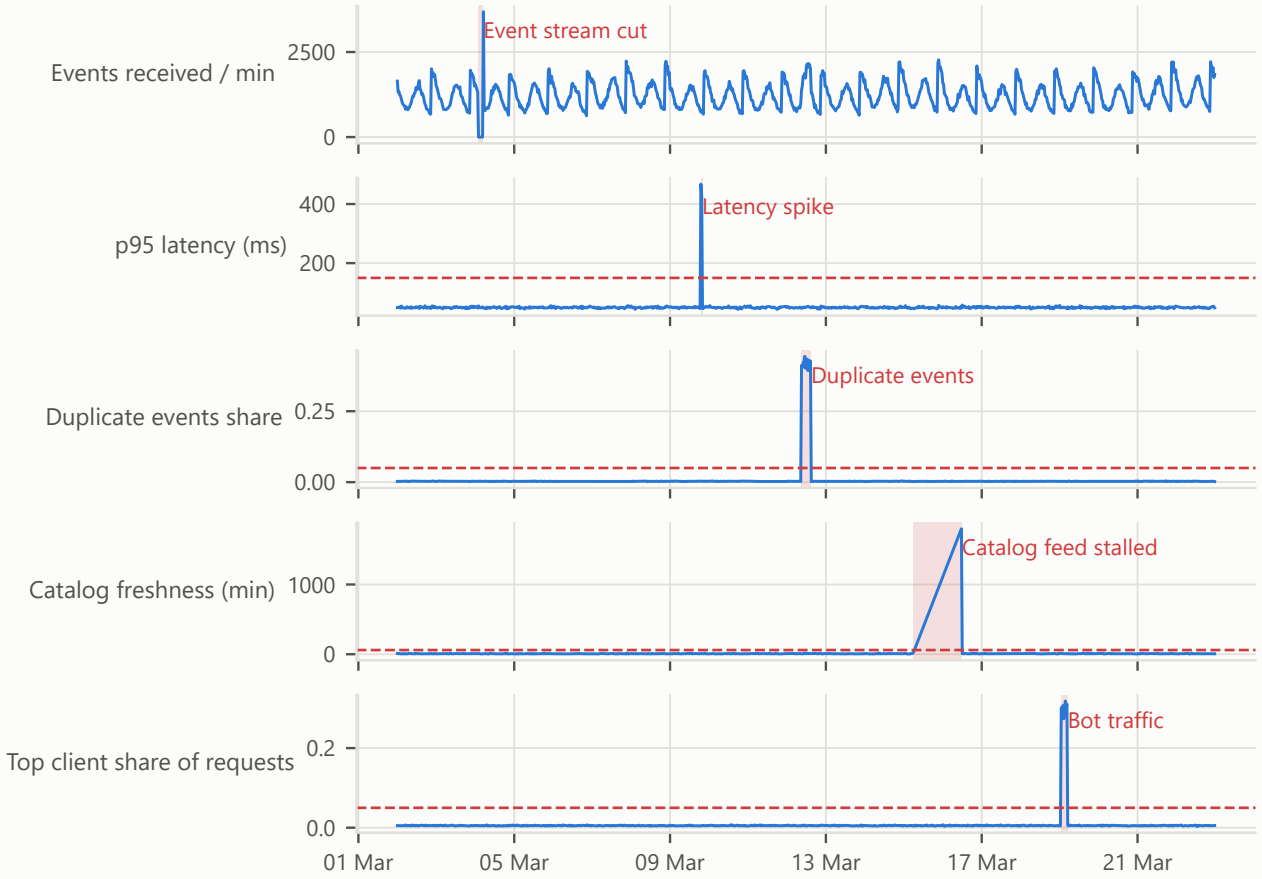
Data: RetailRocket e-commerce dataset (public, CC BY-NC-SA 4.0), 2,756,101 events, May-Sep 2015 · code: sql/, src/adequacy/ · src/ab_design.py · Independent case study based on public data · not affiliated with or endorsed by Albatross AI

Monitoring caught all 5 injected incidents; the catalog-stall rule needs a tighter threshold

SIMULATED

Health is read daily during the live test: if events stop, latency spikes or the catalog freezes, the A/B result is not trustworthy, whatever it says. Interactive version: docs/dashboard.html.

Weeks 5-7 live monitoring (30-min view; red band = injected incident, dashed = alert threshold)



5 of 5 injected incidents detected, 0 false alarms

Sev	Incident	Alert	Delay	Owner
SEV2	Event stream cut	04 Mar 02:10	10 min	Customer engineering lead
SEV1	Latency spike	09 Mar 19:05	5 min	Albatross on-call engineer
SEV3	Duplicate events	12 Mar 09:10	10 min	Customer engineering lead
SEV2	Catalog feed stalled	15 Mar 07:10	70 min	Customer engineering lead + Albatross data engineer
SEV3	Bot traffic	19 Mar 01:10	10 min	Deployment Strategist (resolve)

Delay = from the true start of the problem to the alert. The catalog stall alerted after 70 min because the rule waits for 60 min of staleness: lower it to 30 min.

Normal operating range (median over 3 weeks)

p50 / p95 latency **18 / 46 ms** · errors **0.05%** · fallback **0.6%** · event lag p95 **1.9 s** · new listing live in **4 min** · catalog coverage **56%** → **74%**

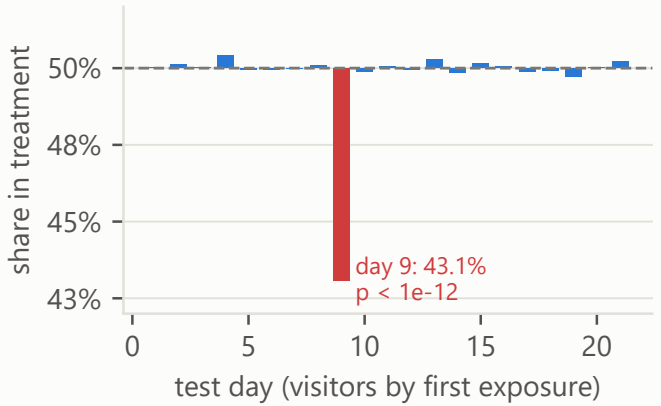
Each alert has an owner and an action in the escalation grid (page 9). Event-stream and bot windows are excluded from A/B metrics for both arms.

Engagement +20% (+15% once novelty fades), conversion +3.3%: go to production, with a holdout

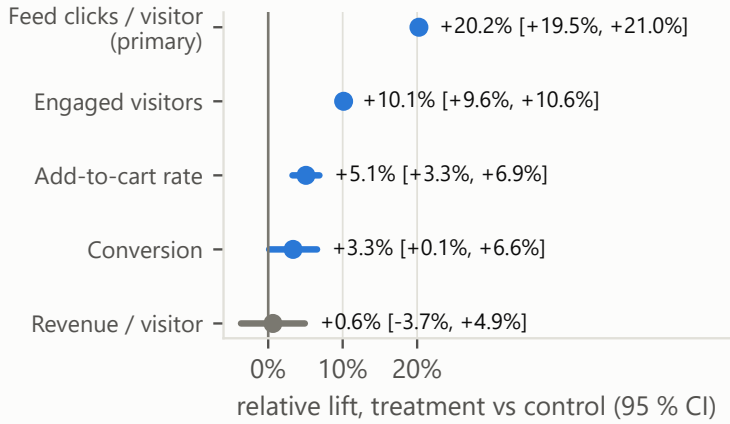
SIMULATED

Read in the order of the analysis plan: trust the split first, then the effect, then how it changes over time and across segments. Simulated effects are deliberately set far below the +84 % to +119 % engagement lifts Albatross's customers published.

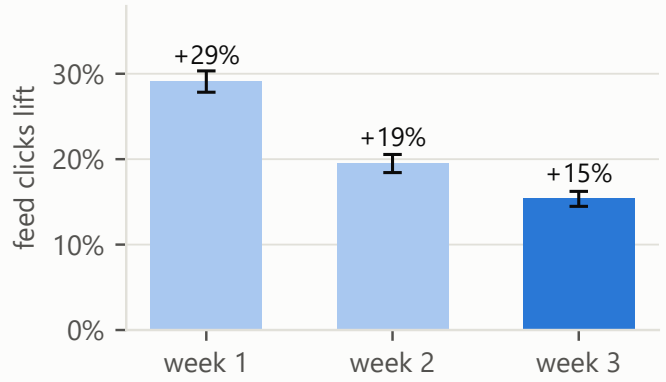
1 · Sample-ratio mismatch found and fixed first



2 · Effects with 95 % confidence intervals



3 · Novelty: the lift fades



Week 3 (+15%) is the steady-state estimate, not the 3-week average (+20%).

4 · Segments (Holm-adjusted p)

Segment	Metric	Lift	95 % CI	p adj.
new	clicks	+16.1%	[+15.1%, +17.2%]	0.00
new	conversion	+4.0%	[-0.8%, +8.8%]	0.30
returning	clicks	+23.0%	[+22.0%, +24.0%]	0.00
returning	conversion	+2.8%	[-1.5%, +7.1%]	0.40
desktop	clicks	+20.0%	[+18.7%, +21.3%]	0.00
desktop	conversion	+3.1%	[-2.8%, +9.0%]	0.40
mobile	clicks	+20.3%	[+19.5%, +21.2%]	0.00
mobile	conversion	+3.4%	[-0.4%, +7.3%]	0.30

Clicks rise in every segment; no segment-level conversion claim survives correction.

5 · Decision: go to production

- ✓ SRM resolved (p >= 0.001 after documented exclusion)
- ✓ Primary metric up, p < 0.05
- ✓ Guardrails within margin
- ✓ Conversion not harmed (CI low > -2 %)

Guardrails: page load +5.9 ms, bounce -1.2 pt. **Condition:** keep a 5 % holdout for 3 months to confirm revenue per visitor.

Guardrails and revenue

Guardrail	Control	Treatment	Max allowed
Page load time	872 ms	878 ms	+50 ms
Bounce rate	38.0%	36.8%	+0.5 pt

Revenue per visitor +0.6% [-3.7%, +4.9%]: inconclusive, not negative. The business case uses the conversion lift; a holdout after launch measures revenue directly.

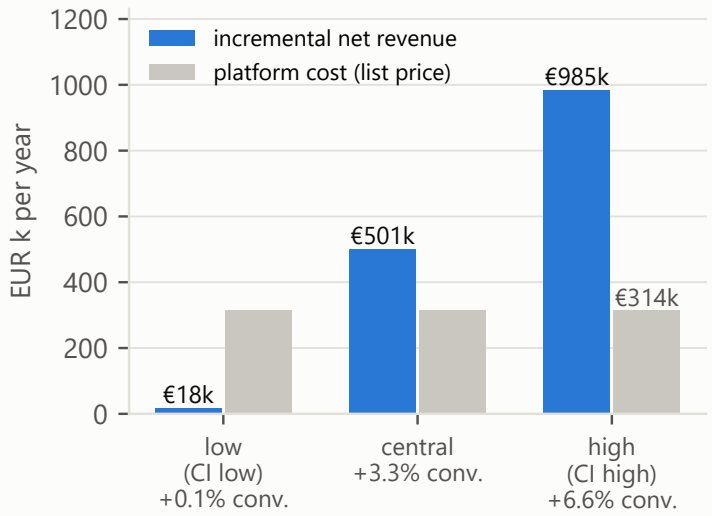
Bootstrap CI (500 resamples) on clicks: [+19.6%, +21.0%]; on revenue: [-3.4%, +4.9%]. Order values vary a lot (average order value -2.6%, p = 0.07).

At list price, the feed pays back above +2.1% conversion; the central case returns 1.6× its cost

SIMULATED + PUBLIC PRICING

Sales support in numbers: the A/B confidence interval becomes a low / central / high case, the public price list gives the cost, and the break-even lift is the number to anchor the negotiation on.

Annual value vs cost at list price



Case	Conv. lift	Extra GMV	Net revenue	After cost	ROI
low	+0.1%	€0.1M	€18k	−€296k	0.1×
central	+3.3%	€4.2M	€501k	+€187k	1.6×
high	+6.6%	€8.2M	€985k	+€671k	3.1×

Cost, from the public price list

GROWTH tier (1-5M MAU): **€20,000** / month · 12.3M predictions / month at €0.50 per 1k: **€6,167** / month · total **€314k** / year.

Revenue side (SIMULATED customer): GMV €250M / year, 50% from homepage visitors, take rate 12%. Lift = conversion lift (order value assumed flat, see page 7). Payback of the €20k one-off integration effort: 1.3 months in the central case.

Sensitivity: net benefit after cost, € k / year

Conv. lift	take 8%	take 12%	take 15%
+1.0%	-214	-164	-127
+2.0%	-114	-14	+61
+3.3%	+16	+181	+305
+5.0%	+186	+436	+623
+6.6%	+346	+676	+923

One-pager for the negotiation

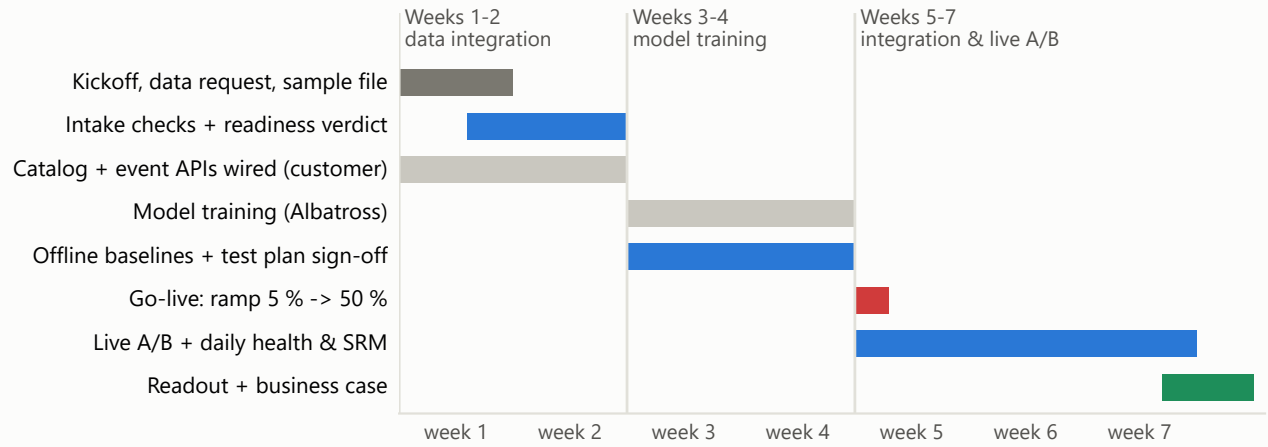
- **Break-even:** +2.1% conversion on the homepage feed covers list price.
- **Measured:** +3.3% conversion, +15% steady-state engagement, guardrails clean.
- **Central case:** €4.2M extra GMV, €187k net per year after cost (1.6×).
- **Risk:** the CI lower bound (+0.1%) does not pay; a 5 % holdout measured at month 3 gives both sides a shared success metric.
- **Upside not counted:** engagement value, seller retention, more surfaces (Similar Items: +10 % platform fee).

SIMULATED data (src/simulate.py, seed 42), calibrated on the RetailRocket base rates and kept below Albatross's published lifts; assumptions in data/ASSUMPTIONS.md · pricing: usealbatross.ai/pricing, collected 2026-11-11 · src/business_case.py · Independent case study based on public data · not affiliated with or endorsed by Albatross AI

Two people on the customer side, one owner on ours: who does what, when, and when to escalate

The customer relationship is the delivery mechanism: clear asks with formats and dates, one weekly report with the same structure, and written rules for what is fixed quietly and what is escalated.

Who does what, week by week



	Customer engineering lead	Customer PM	Deployment Strategist
Wk 1-2	Wire Catalog + Event APIs, history drop	Owns dates, unblocks access	Data request, intake report, readiness verdict
Wk 3-4	Fix intake conditions, exposure logging	Signs the analysis plan	Baselines, power, plan, go-live checklist
Wk 5-7	Feed integration on the homepage	Weekly check-in, readout	Daily health + SRM, weekly report, readout + business case

Escalate or resolve? (excerpt of playbook/04)

Signal	Rule	Who
p95 latency	> 150 ms for 10 min	Escalate now: on-call
Event stream down	< 20 % of yesterday, 15 min	Escalate: customer engineer
Catalog stalled	freshness > 30 min	Escalate: both data owners
SRM	p < 0.001	Investigate, then data scientist
Duplicates, bots	> 5 %	Resolve, report weekly
Data drop late	2 working days	Follow-up; PM after 5 days

Resolve alone what is reversible and invisible to users; escalate early, with facts, when users, revenue, the timeline or trust are at stake.

Templates in playbook/

```
Subject: Evaluation – data drop: 2 fixes needed to keep
the week-3 training date

1. 9.3% of events point to items missing from
   the catalog (49815 ids). Could you re-export
   the catalog including sold and delisted listings?
2. Time zone: please confirm timestamps are UTC.

Impact on the plan: if we have it by [date], training
still starts on [week-3 date].
```

01 data request · 02 follow-up · 03 weekly report · 04 escalation grid · 05 go-live checklist

Methods, limits and AI note

Every number in this report comes from a query or a script in the repository; simulated numbers are labelled as such on every page.

Sources and licences

- **Albatross public pages** (home, technology, evaluation, primitives, customers, Wallapop case study, pricing, company, resources), collected 2026-11-11, one request every 3 s, robots.txt respected. Facts table with URL and date: data/public/albatross_facts.csv. Page copies are not redistributed (site terms).
- **RetailRocket e-commerce dataset**, CC BY-NC-SA 4.0. Integrity checked against the published totals (2,756,101 events, 1,407,580 visitors, 22,457 transactions) and SHA-256 recorded. Raw files are not redistributed; only aggregates.
- **Simulated data**: src/simulate.py, seed 42; every assumption in data/ASSUMPTIONS.md.

Methods

- SQL (DuckDB): intake checks, filtering rules, sessions (30-min gap), adequacy metrics, co-visitation, time-split evaluation.
- Python: readiness scoring, sample size (two-proportion and mean formulas), Welch z-tests, bootstrap, chi-square SRM, Holm correction, rule-based health alerts, business-case sensitivity.
- Toolkit: `python -m adequacy check --events … --catalog …`, 8 pytest tests on a synthetic drop with injected anomalies.

AI note

> *AI-assisted workflow: I used Claude, mainly to write scripts faster. I drove the project: I defined the questions, chose and validated the data sources, made the analytical decisions and checked every result.*

Limits

- RetailRocket is a 2015 e-commerce shop, not a 2026 second-hand marketplace: smaller, shorter sessions, hashed item properties.
- Weeks 5-7 are simulated: they demonstrate the checks and the reasoning, not real Albatross or customer performance.
- The baselines are deliberately simple and are not Albatross's models; offline recall does not predict online lift.
- Readiness thresholds and alert rules are this playbook's choices, written down to be challenged.
- Business case: customer GMV, take rate and homepage share are assumptions; only the pricing is public.

Reproduce

```
pip install -r requirements.txt && pip install -e .
python src/load_dataset.py   # needs the raw dataset
python src/run_intake.py
python src/baselines.py
python src/ab_design.py
python src/simulate.py
python src/ab_readout.py
python src/health_monitor.py
python src/business_case.py
python src/build_dashboard.py
python src/build_report.py
pytest
```